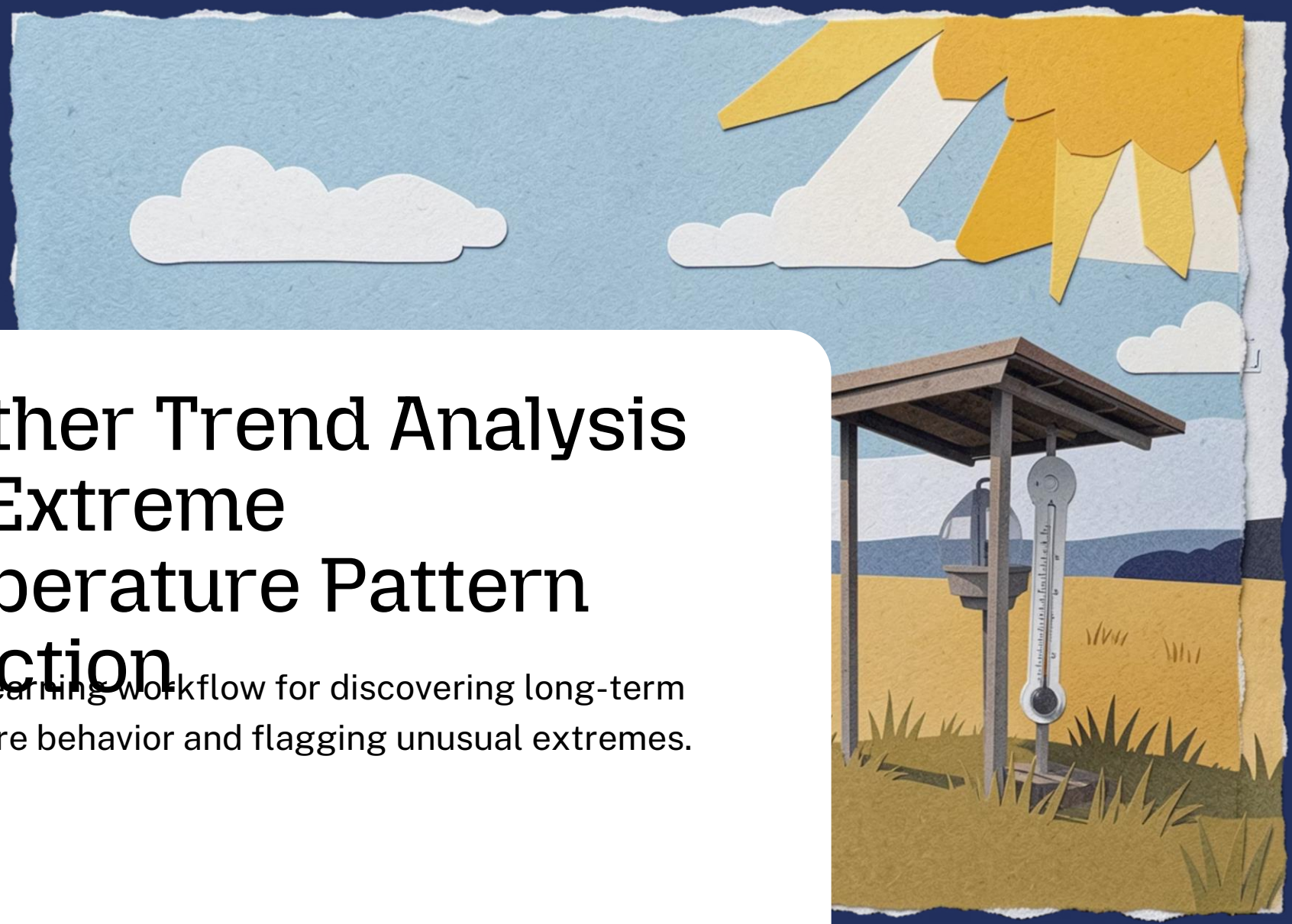


Weather Trend Analysis and Extreme Temperature Pattern Detection

Machine learning workflow for discovering long-term temperature behavior and flagging unusual extremes.

roll1
roll2
roll3
roll4

Made with GAMMA

Problem Statement



Problem

Daily weather records contain noise, missing values, seasonality, and station-level variation.



Need

Combine trend analysis with a repeatable detector for unusually hot or cold observations.



Project aim

Transform raw observations into interpretable trends, engineered signals, and evaluated predictions.

Data Foundation

Chosen dataset: NOAA Global Historical Climatology Network Daily (GHCN-Daily), a quality-reviewed archive of daily land-station climate summaries.

Available inputs: daily maximum temperature, minimum temperature, precipitation, snowfall, snow depth, station metadata, and observation dates.

Project scope: select one station or city subset and define the analysis window before model training.

Target: classify an observation as normal vs. extreme, with optional hot/cold subclasses.

FEATURE ENGINEERING

Feature engineering turns weather history into learnable signals

01

Clean

Handle missing and flagged observations; standardize units and align dates.

02

Calendar

Create day-of-year, month, season, and cyclical sine/cosine signals.

03

Rolling

Build 3-day and 7-day means, volatility, and lagged temperature values.

04

Anomaly

Compare observations with a seasonal baseline and apply the project

Use a time-aware train, validation, and test split so future information never leaks into the past.

Raw observations → aligned dates → temporal context → anomaly signal

BASELINE MODEL

A transparent baseline makes the detector easy to evaluate



Baseline: Logistic regression or random forest for extreme-event classification.

Inputs: Lagged temperature, rolling statistics, precipitation context, seasonality, and station metadata.

Output: Class label plus probability score for each observation. **Interpretability:** Feature importance or coefficient analysis explains which signals drive the flag.

Decision rule: Compare with one stronger model only if recall improves without sacrificing interpretability.

EVALUATION

Outputs should report more than accuracy

Classification

Accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.

Trend analysis

Rolling average, seasonal baseline, anomaly score, and event frequency over time.

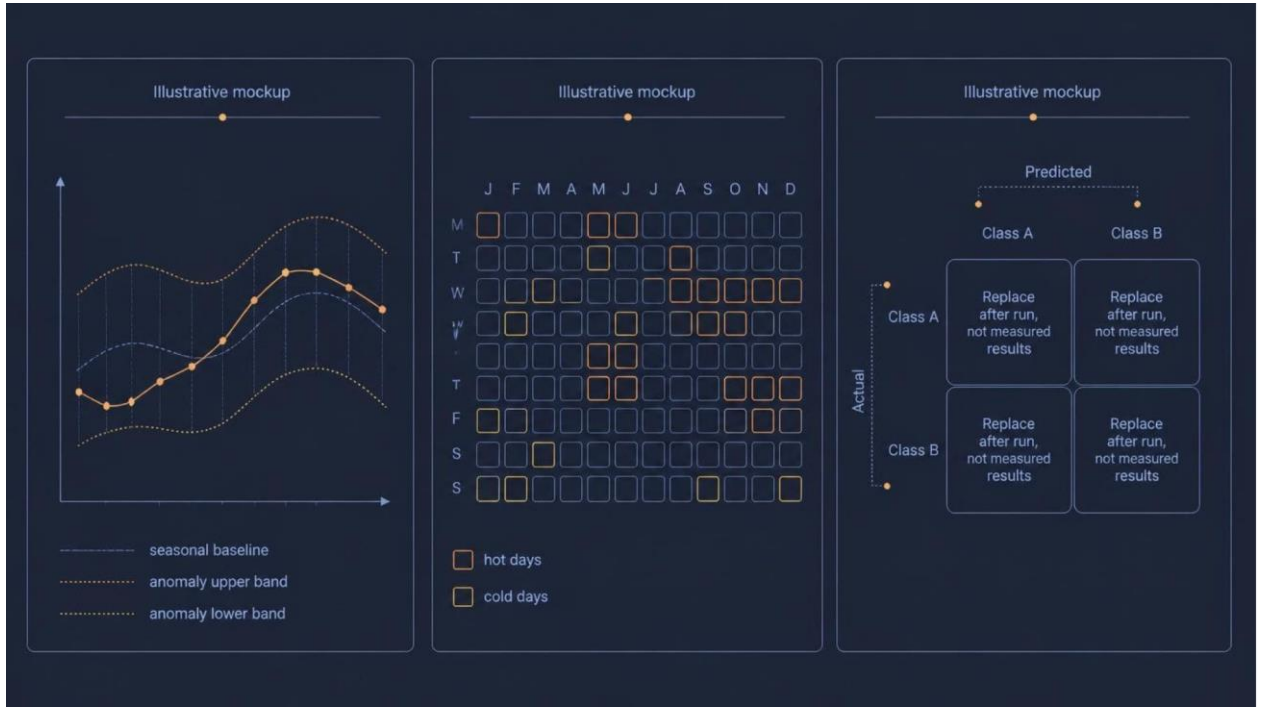
Operational view

Predicted probability, alert threshold, false-positive review, and seasonal error breakdown.

A useful detector explains what it flags, when it fails, and how stable the signal is

RESULTS AFTER TRAINING

Results: connect the trend line to the extreme-event signal



Metrics

Accuracy [value]
Precision [value]
Recall [value]
F1-score [value]
ROC-AUC [value]

Readout

Check whether hot or cold events are missed more often, and which seasons create the most false alarms.

NEXT STEPS

Future goals: move from detection to dependable forecasting



Broaden the comparison

Extend from one station to multi-station and regional comparisons.



Strengthen the detector

Test calibrated thresholds, class balancing, and stronger time-series models.



Monitor reliability

Add explainability, uncertainty bands, and drift monitoring for new observations.



Package the workflow

Turn the analysis into a reproducible dashboard or alerting tool.

Thank you